

Further Investigation into Regime Detection Neural Network Signal for Trading Strategy in Equity, FX and Rates Futures

Author: Dr. Tim Delpont
Email: delponttim@gmail.com

Executive Summary

Further investigation into the potential use of the neural network-provided regime detection signal has been conducted following response from the signal's provider which highlighted where advantageous execution structures might lie. The provider specifically encouraged a 'per trade' analysis of the signal's hit rate instead of a daily settlement. This encouragement was grounded in their observation that each trade's value was historically realising returns during overnight and multi-day holding periods - meaning a daily scored hit-rate could punish its 'early' entry record. It was suggested that using further execution refinement (i.e. intra-day entry and volume) may develop additional edge, however this has not been explored as of yet. Removing daily settlement improved the general outlook of the signals performance, serving to smooth apparent volatilities and give the appearance of improved consistency/hit-rate in comparison to previously reported results. Each of the signals showed relatively strong correlation with market returns. Only one signal provided truly consistent performance across three quarters (ticker "ZN", U.S. 10-year Treasury Note). It's accuracy in predicting bearish trends being the main contributor to its success, alongside fairly correlated behaviour when aggregating its three quarters of auditable history.

Introduction

Please refer to previous report for an introduction of the neural network (NN) and relevant tradeable assets being assessed.

Method

The following investigative steps will be taken to address provider suggestions:

1. Compile data and process into 'per trade' records for assessment.
2. Assess the general performance of the signal under new framework and establish correlation with market
3. Assess the long/short performance of the signal.
4. Make observations and recommendations

Analysis

Figure 1 provides a summary of trade performance by ticker and quarter. This details how each signal results of trade lengths ranging between 1 and 27 days, frequently averaging periods of 2-17 days. Returns appear move stable despite some assets making loses across the quarter, as evident by less extreme swings in Sharpe ratios. Market correlations across the assets appear relatively high. This is consistent with previous findings that highlighted long-only positions in the underlying could potentially outperform the signal.

Ticker "ZN" (U.S. 10-year Treasury Note) is the strongest and most consistent signal, with a Sharpe between 2 and 3 and returns of 1.4-5.7% quarter. It's aggregated trades over the available three quarters of auditable records results in relatively uncorrelated returns.

Figure 2 aggregates the performance of each ticker across each quarter of records and summarises their performance under bullish and bearish regime detections. A particular interest is given to performance in bearish conditions as a proxy for signal 'alpha' given the generally accepted positive (bullish) drift of selected assets. Ticker

Ticker	Quarter	Trades	Mean_Return	Volatility	Mean_Log_Return	Hit_Rate	Avg_Length	Min_Length	Max_Length	Sharpe	Max_Drawdown	Total_Return	Market_Corr
6E	2025Q3	6	0.0014	0.0094	0.0013	16.67%	9.67	2	25	1.08	-0.010	1.008	0.99
6E	2025Q4	9	-0.0022	0.0069	-0.0022	33.33%	5.89	1	25	-2.33	-0.015	0.988	0.71
6E	2026Q1	5	-0.0074	0.0146	-0.0075	40.00%	10.60	1	19	-3.67	-0.044	0.952	0.95
6E	2026Q2	8	0.0024	0.0085	0.0024	50.00%	6.00	1	17	1.96	-0.011	0.970	0.41
ES	2025Q3	5	0.0270	0.0225	0.0264	100.00%	12.40	1	39	9.50	0.000	1.141	1.00
ES	2025Q4	9	0.0043	0.0155	0.0042	55.56%	6.33	1	29	2.07	-0.025	1.185	-0.23
ES	2026Q1	7	-0.0101	0.0174	-0.0103	42.86%	8.29	2	14	-4.44	-0.085	1.103	-0.02
ES	2026Q2	16	0.0038	0.0242	0.0035	37.50%	3.44	1	14	1.10	-0.075	1.167	0.68
FGBM	2025Q3	3	-0.0027	0.0027	-0.0028	0.00%	17.67	1	27	-7.52	-0.003	0.992	0.81
NQ	2026Q1	8	-0.0060	0.0106	-0.0061	37.50%	7.25	1	21	-4.31	-0.054	0.953	0.76
NQ	2026Q2	21	0.0113	0.0231	0.0110	76.19%	2.67	1	8	3.65	-0.025	1.201	0.71
ZN	2025Q4	8	0.0018	0.0045	0.0017	75.00%	7.12	2	15	2.93	-0.004	1.014	0.03
ZN	2026Q1	7	0.0018	0.0064	0.0017	57.14%	8.29	1	20	2.07	-0.006	1.026	0.76
ZN	2026Q2	18	0.0016	0.0046	0.0016	66.67%	3.11	1	9	2.63	-0.006	1.057	-0.70

Figure 1: Summary of signal performance by ticker and quarter

"ZN" is highlighted for having a strong 'short' hit-rate, contributing to its overall trading success. Remaining tickers have relatively poor hit-rates or are accompanied by negative returns meaning the signal missed crucial opportunities.

Ticker	Type	Trades	Mean_Return	Volatility	Mean_Log_Return	Hit_Rate	Avg_Length	Min_Length	Max_Length	Sharpe
6E	long	15	-0.0015	0.0127	-0.0016	40.00%	9.27	1	25	-1.49
6E	short	13	-0.0005	0.0047	-0.0005	30.77%	5.62	1	25	-0.87
ES	long	19	0.0103	0.0274	0.0099	57.89%	9.58	1	39	4.96
ES	short	18	-0.0018	0.0147	-0.0019	44.44%	2.78	1	11	-0.90
FGBM	long	2	-0.0031	0.0037	-0.0031	0.00%	26.00	25	27	-6.10
FGBM	short	1	-0.0020	nan	-0.0020	0.00%	1.00	1	1	nan
NQ	long	15	0.0131	0.0272	0.0127	66.67%	6.00	1	21	4.51
NQ	short	14	-0.0004	0.0108	-0.0005	64.29%	1.71	1	5	-0.22
ZN	long	17	0.0013	0.0047	0.0012	58.82%	4.71	1	13	2.39
ZN	short	16	0.0021	0.0051	0.0021	75.00%	5.69	1	20	3.96

Figure 2: Summary of signal performance in short/long regimes

Summary of Findings

Each signal continues to have potentially positive trading strategies under the right execution logic but ticker "ZN" appears to naturally outperform others. It is therefore more likely to retain its positive attributes and be successfully executed when further refined for deployment (i.e. volume and timing).

It is not clear yet whether the nature of the provided signal suits and avoid conflict with the portfolio enacted by [redacted] and thus it is recommended that further investigation is suspended until [redacted]'s senior quantitative traders can review present results.

Further Investigations

No further investigations are detailed at present.